

New Horizons

The first mission to Pluto and the Kuiper Belt



Pluto in enhanced color (New Horizons data)

INTRODUCTION

New Horizons reached Pluto, a world no spacecraft had visited

Goal: map Pluto and its moons, then explore the Kuiper Belt.

9.5 years

from launch to Pluto

4.9 billion

km traveled (3 billion miles)

1st

spacecraft to visit Pluto

Outline

1

Why it was built

2

The spacecraft

3

The journey and findings

4

Conclusion and impact

WHY IT WAS BUILT

Pluto was the last unexplored world of the old nine planets



Total unknown

Telescopes showed only a blurry dot.



Closing window

Pluto's thin atmosphere may freeze as it moves away from the Sun.



Kuiper Belt

An icy belt found in 1992 that no spacecraft had visited.

A piano-sized spacecraft that runs on plutonium power



Artist's concept: New Horizons at Pluto and Charon



Energy

One plutonium RTG gives about 200 W at Pluto, where sunlight is 1/1000 of Earth's.



Communication

A 2.1 m dish sends data at about 1 kilobit per second. One-way signals take 4.5 hours.



Computers and instruments

Two backup 12 MHz computers run 7 instruments, including a student-built dust counter.

After 9.5 years, New Horizons found a geologically active Pluto

Distance from the Sun (AU)

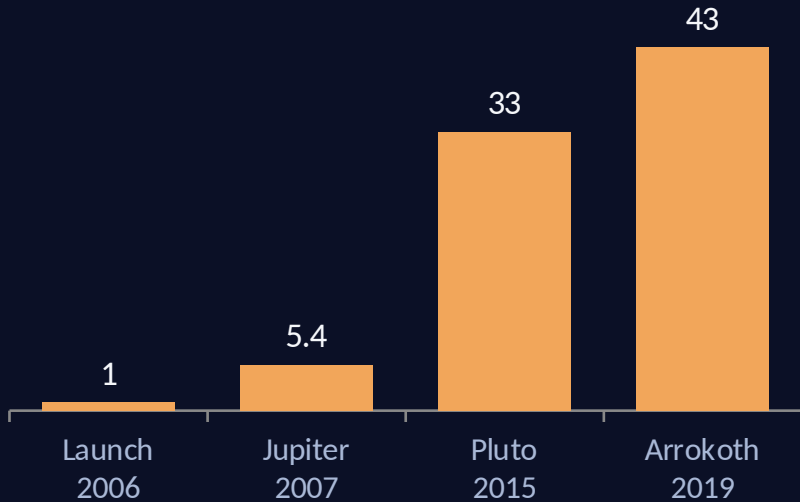


Illustration of Pluto's heart-shaped icy plain

12,500 km
closest approach

3,500 m
ice mountains

5 moons
found around Pluto

Conclusion



Common thread

Curiosity about places humans cannot reach drives better machines.



Major influences

Stronger rockets, nuclear power, tough electronics, and long-range radio.



Cause and effect

Questions drive missions. Missions bring data that raise new questions.

Impact

The flyby was broadcast live, and its data took 16 months to arrive. New Horizons passed Arrokoth in 2019 and still works more than 60 AU from the Sun, in the same NASA program as Juno and Dragonfly.